

20.07.2026

PRACTICAL-1

BASIC NETWORKING

COMMANDS:

AIM:

Study of various Network commands used in Linux and windows.

COMMANDS:

1.) arp-a :-

ARP is short form of address resolution protocol. It will show the IP address of your computer along with the IP address and MAC address of your router.

OUTPUT:

172.16.8.149	00:27:0e:13:f3:fe	enp3s0
172.16.9.212	7c:57:58:38:da:a3	enp3s0
172.16.8.186	7c:57:58:38:c6:24	enp3s0
172.16.8.249	00:27:0e:13:f5:77	enp3s0
172.16.11.207	7c:57:58:38:46:87	enp3s0

2.) hostname:

This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

OUTPUT:

localhost.localdomain

3.) netstat: (network statistics)

netstat displays a variety of statistics about a computer's active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics, etc.

Eg:- netstat -n

OUTPUT:

Proto	Local Address	Foreign Address	State
tcp	localhost.localdomain:ssh	172.16.8.135:56449	ESTABLISHED
tcp	localhost.localdomain:ssh	172.16.11.207:59129	ESTABLISHED
tcp	localhost.localdomain:ssh	172.16.8.140:52171	ESTABLISHED
tcp	localhost.localdomain:ssh	172.16.8.166:62861	ESTABLISHED
tcp	localhost.localdomain:ssh	172.16.8.162:50007	ESTABLISHED

4) nslookup: (name server lookup)

is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain, nslookup can operate in two modes: interactive and non-interactive.

Eg:- nslookup www.google.com

OUTPUT:

Server : 172.16.8.1

Address : 172.16.8.1#53

Non-authoritative answer:

Name: www.google.com

Address: 142.251.156.119

Name: www.google.com

Address: 142.251.152.119

Name: www.google.com

Address: 142.251.154.119

5) ping: (Packet INternet Grooper)

Command is the best way to test connectivity between two nodes. Ping uses ICMP (Internet Control Message Protocol) to communicate to other devices.

1) # ping hostname (ping localhost)

2) # ping ip address (ping 4.2.2.2)

3) # ping fully qualified domain name (ping www.facebook.com)

OUTPUT:

ping www.google.com

PING www.google.com (142.251.155.119) 5(84) bytes of data:

64 bytes from 142.251.155.119 (142.251.155.119): icmp_seq=1 ttl=119 time=3.67 ms
64 bytes from 142.251.155.119 (142.251.155.119): icmp_seq=2 ttl=119 time=3.87 ms
64 bytes from 142.251.155.119 (142.251.155.119): icmp_seq=3 ttl=119 time=5.29 ms
64 bytes from 142.251.155.119 (142.251.155.119): icmp_seq=4 ttl=119 time=3.35 ms
64 bytes from 142.251.155.119 (142.251.155.119): icmp_seq=5 ttl=119 time=3.43 ms

--- www.google.com ping statistics ---

5 packets transmitted, 5 received, 0% packet loss, time 4005 ms.

rtt min/avg/max/mdev = 3.351/3.928/5.299/0.710 ms.

6.)

Route :

route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an internet.

OUTPUT:

Kernel	IP routing table
Destination	Gateway Genmask Flags Metric Ref Use Iface
default	gateway 0.0.0.0 UG 100 0 0 emp350
172.16.8.0	0.0.0.0 255.255.252.0 U 100 0 0 emp350

7.)

ipconfig/all:

This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system.

OUTPUT:

Windows IP configuration

Ethernet adapter Ethernet:

Connection-specific DNS suffix . . . :

Link-local IPv6 Address : fe80::36c4:6b74:7539:e2f8%4

IPv4 Address. : 192.168.56.1

Subnet Mask : 255.255.255.0

Default Gateway :

8.)

nbtstat -a:

This command helps solve problems with NetBIOS name resolution.

OUTPUT:

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP).

LINUX NETWORKING COMMANDS:

1. ip:

The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems. The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels.

ip <OPTIONS> <OBJECT> <COMMAND>

ip address show

OUTPUT:

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue
State UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

valid-lft forever preferred-lft forever

inet ::1/128 scope host

valid-lft forever preferred-lft forever

2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000

link/ether 08:00:27:ff:fe:56:3b:04 brd ff:ff:ff:ff:ff:ff

inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic

noprefixroute enp0s3 valid-lft 8634 sec preferred-lft 8634 sec

inet 6:fe80::a00:27ff:fe56:3b04/64 scope link noprefixroute

valid-lft forever preferred-lft forever

2. ifconfig:

The ifconfig command was/is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed above.

OUTPUT:

enp0s3: flags=4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500

inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255

inet fe80::a00:27ff:fe56:3b04 prefixlen 64 scopeid 0x20<link>

3. mtr

MTR (Mett's traceroute) is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute commands.

OUTPUT:-

local host, local domain (:::1) → localhost

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Keys: ~~h~~ Help Display mode ~~R~~ Restart Statistics ~~O~~ Order of details ~~q~~ quit

Host	loss%	Packets		Avg	Pings			StDev
		Snt	last		Best	Worst	StDev	
1.) localhost	0.0%	95	0.1	0.1	0.1	0.7	0.1	

mtr google.com

Host	loss%	Packets		Avg	Pings			StDev
		Snt	last		Best	Worst	StDev	
1.) gateway	0.0%	44	2.2	1.7	0.5	2.8	0.4	

waiting for reply

11.) prnma-be-in-f14.ler100.net	3.1	96	411.6	437.2	336.8	573.5	34.2
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4) tcpdump

The tcpdump command is designed for capturing and displaying packets.

#dnf install -y tcpdump

OUTPUT:

tcpdump: verbose output suppressed, use -v[-v]... for full protocol decode listening on enp0s3, link-type EN10MB (Ethernet)
 Snapshot length 26144 bytes 10:41:42.649924 IP ftp.yz.yamagata-u.ac.
 IP: http → localhost.localdomain.59614: F

5) ping :

ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo request messages.

OUTPUT:

PING google.com (142.250.134.102) 56(84) bytes of data:

9.) pathping:

Pathping is unique to Windows and is basically a combination of the ping and Tracert commands. Pathping traces the route to the destination commands.

OUTPUT:

Usage: pathping [-g host-list] [-h maximum hops] [-i address] [-n
[-p period] [-q num-queries] [-w timeout]
[-4] [-6] target-name.

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RESULT:-

Thus the various network commands used in Linux and Windows was executed successfully.